

Otsu Thresholding

In the lecture, we saw how the function 'otsuthresh' can be used to determine a normalized intensity threshold for an image.

Write a function **OtsuThreshold**, which takes as input an [MxN] uint8 image matrix **img**, and outputs the Otsu thresholded binary image **msk** and integer threshold **thrsh** used to make **msk**.

The steps are:

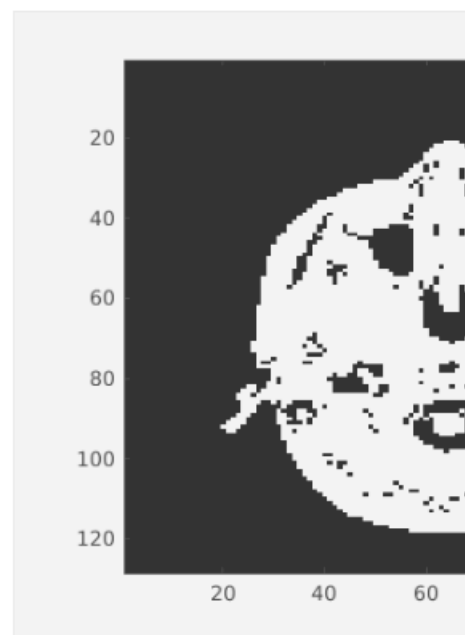
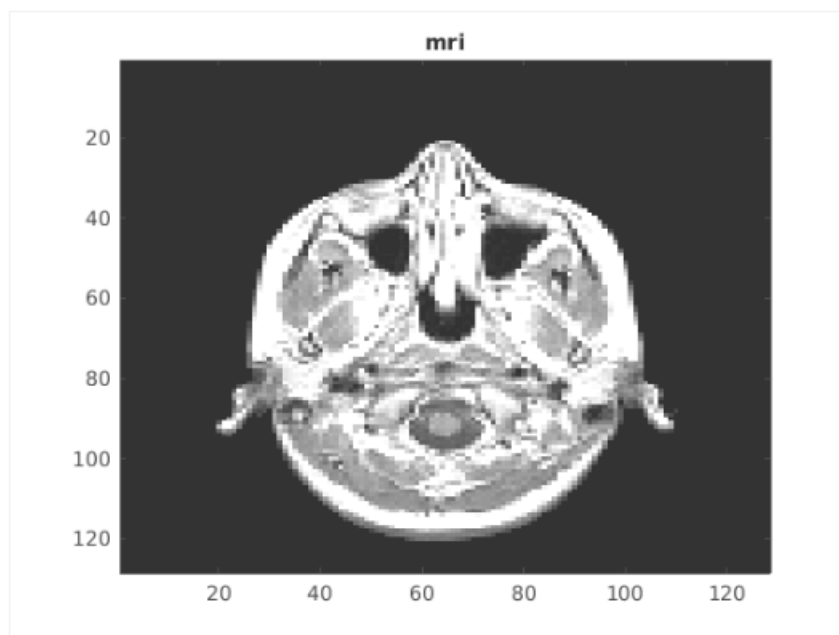
- Define the Otsu threshold **thrsh** using the histogram of **img**.
- Apply the threshold to **img** to make **msk**.

Running the example code below, your output should look identical to this:

Output

thrsh =

36



Function ?

 Save

 Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 function [msk,thrsh] = OtsuThreshold(img)
2 % Define the Otsu threshold 'thrsh' using the histogram of img
3 thrsh = 255*otsuthresh(imhist(img));
4 % Apply the threshold to 'img' to make 'msk'
5 msk = img > thrsh;
6 end
```

Code to call your function ?

 Reset

```
1
2
3
4
5
6
7
```

```
8 img = imread('mri.tif');  
9 [msk,thrsh] = OtsuThreshold(img);  
10
```

[▶ Run Function](#)

Assessment: All Tests Passed

[Submit](#)

✔ Is output correct with random valued input

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